

PHASE 1.1 CONSOLIDATED WORK REPORT

District Baseline Research Methodology

A work report documenting the goal, source authority, cleanup decisions, validation gates, discovery protocol, deduplication logic, and research limitations behind the district-only Shiva temple discovery baseline.

Boundary statement: this is a discovery baseline built from official district data and Google Places observations. It is not an exact count of all Shiva temples and not an official temple census.

ACTIVE LGD DISTRICTS

785

District-only administrative baseline

SEARCH TASKS

7,065

9 Shiva-related queries per district

UNIQUE CANDIDATES

74,117

Current district-attributed Google Place IDs

1. Goal And Research Frame

Goal Of Phase 1.1

Phase 1.1 establishes a clean, national, district-level discovery baseline for likely Shiva temple candidates in India. The purpose is to create a comparable administrative starting point before adding towns, urban local bodies, sub-districts, or villages.

The working research question is: **using official district geography and a bounded Google Places discovery protocol, where do Shiva-related temple candidates appear, and how strong is the name-based Shiva signal?**

PREPARED LGD ROWS

785

Normalized from LGD district source

DISTRICTS WITH LGD CODES

785

Active district rows carrying LGD district code

INACTIVE AUDIT ROWS

1

Retained for history, excluded from Phase 1.1

Why District-Only First

- Districts are official, recognizable units for research and government communication.
- A district layer prevents early mixing of towns, ULBs, tehsils, villages, and legacy pilot rows.
- The task volume is controlled: 785 districts times 9 keywords.
- It creates a national benchmark before deeper local expansion.

What This Work Can Support

- District-level prioritization for manual review.
- Evidence packs for academic or government conversations.
- A repeatable pipeline for later location layers.
- An audit trail for administrative naming decisions.

2. Source Authority

Source Of Truth Decisions

ENTITY	PHASE 1.1 SOURCE	HOW IT WAS TREATED
District names and codes	Data.gov.in LGD district data from Ministry of Panchayati Raj	Administrative source of truth for district rows and LGD district codes.
Location hierarchy	LGD district/state columns	Normalized into india_locations with active district scope.
Temple candidate names	Google Places displayName	Discovery observation only, not canonical temple name truth.
Unique candidate identity	Google Place ID	Operational dedupe key for API results.
Shiva relevance	Configured high and medium Shiva term lexicon	Name-based confidence classification for triage and review.

District Names

The district layer uses Local Government Directory data as the administrative reference. The Ministry of Panchayati Raj describes LGD as a directory for land/revenue regions and local governments, with unique LGD codes assigned to government bodies and administrative entities.

<https://panchayat.gov.in/en/lgd/>

Temple Names

There is no final temple-name source of truth in Phase 1.1. Google Places names are treated as discovered field signals. A future verification phase should triangulate candidates against temple trusts, state religious/endowment departments, gazetteers, tourism boards, ASI/state archaeology sources, local records, and manual review.

External Source Notes

- Data.gov's LGD catalog is attributed to the Ministry of Panchayati Raj and is released through the Open Government Data platform. The catalog page observed on May 12, 2026 showed Published On 22/07/2022 and Updated On 10/05/2026.
- Google Places Text Search returns place objects based on a text query and selected field mask. The API documentation notes that result lists are not guaranteed to be identical for repeated requests and that pagination is bounded.

3. Data Cleanup And Validation

Administrative Baseline Workflow

- 1

Normalize LGD file.
CSV headers and codes were mapped into the project's standard location schema.
- 2

Compare LGD to database.
Districts were compared by LGD code, exact normalized name, canonical name, and high-similarity fuzzy match within the same state.
- 3

Resolve naming differences.
Known administrative spelling and rename variants were captured as canonical aliases, then rechecked.
- 4

Handle manual exceptions.
Bajali and Tamulpur were inserted from LGD. Itanagar Capital Complex was retained inactive as a pilot/audit row outside the LGD district baseline.
- 5

Lock Phase 1.1 scope.
Only active district rows were used for task generation and scoped reporting.

Alignment Funnel

VALIDATION PASS	AUTO	REVIEW	MISSING IN DB	DB ONLY
Initial LGD vs DB review	737	27	21	22
After canonical aliases	783	0	2	1
After manual exceptions	785	0	0	1

Initial Review

STATUS	ROWS
auto	737
db_only	22
missing_in_db	21
review	27

After Aliases

STATUS	ROWS
auto	783
db_only	1
missing_in_db	2

Final Review

STATUS	ROWS
auto	785
db_only	1

4. Discovery Protocol

Search Design

- Each active district received the same keyword set for comparability.
- Queries used the pattern: **keyword in district, state, India**.
- The run used Google Places Text Search with **includedType: hindu_temple**, **strictTypeFiltering: true**, **regionCode: IN**, and **languageCode: en**.
- Fields were limited to the needed result attributes, including ID, display name, address, location, type, and Maps URI.

Keyword Set

KEYWORD
Shiva temple
Shiv Mandir
Mahadev temple
Mahadev Mandir
Shankar Mandir
Someshwar temple
Vishwanath temple
Lingeshwar temple
Rudreshwar temple

DONE TASKS

7,064

100.0% of district tasks

FAILED TASKS

1

Retained for audit and rerun

RAW OCCURRENCES

229,475

Sum of completed district task result counts

Task Completion



● Done 7,064 (100.0%) ● Failed 1 (0.0%)

5. Deduplication And Classification

Dedupe Method

Within each API response, repeated Google Place IDs were collapsed. During database writes, candidates were upserted by `google_place_id`, so rediscovery through another query updated the same candidate instead of creating a second candidate row.

This is an operational dedupe method. It avoids repeated API hits becoming repeated candidate records, but it does not prove canonical cultural identity where Google itself has split or merged place records.

Classification Method

Candidate confidence is based on the discovered name only. High-confidence terms include direct Shiva variants such as Shiva, Shiv, Mahadev, Vishwanath, Somnath, Kedarnath, Omkareshwar, Lingeshwar, linga, and lingam. Medium-confidence terms include Shankar, Ishwar/Eshwar, Nath, Rudra, Someshwar, and Bholenath.

Low confidence means no configured Shiva-specific term matched the discovered name. It does not mean the candidate is not a Shiva temple.

UNIQUE PLACE IDS

74,117

Deduped current district-attributed candidates

DUPLICATES REMOVED

155,358

67.7% of raw occurrences

HIGH + MEDIUM SIGNAL

51,243

69.1% of unique candidates

Confidence Mix



● High 43,877 (59.2%) ● Medium 7,366 (9.9%) ● Low 22,874 (30.9%)

6. Research Findings Snapshot

Top States By Unique Candidate Volume

STATE / UT	UNIQUE	HIGH	HIGH+MEDIUM SHARE
Uttar Pradesh	8,493	5,727	77.9%
Rajasthan	5,345	3,640	72.6%
Bihar	5,241	3,394	74.6%
Maharashtra	4,784	2,936	79.9%
Gujarat	4,341	3,135	78.7%
Tamil Nadu	4,029	962	40.1%
Karnataka	3,704	1,454	74.5%
Odisha	3,371	2,096	70.8%

Top Districts By Unique Candidate Volume

STATE	DISTRICT	UNIQUE	HIGH	HIGH+MEDIUM SHARE
Karnataka	Uttara Kannada	237	107	92.4%
West Bengal	South 24 Parganas	229	144	75.5%
Gujarat	Surendranagar	225	142	79.6%
Bihar	Saran	218	140	79.4%
Uttar Pradesh	Jaunpur	211	144	81.5%
Bihar	Samastipur	209	140	79.9%
Tamil Nadu	Krishnagiri	206	81	67.5%
Andhra Pradesh	West Godavari	205	67	39.0%
Karnataka	Tumakuru	204	72	76.0%
Rajasthan	Nagaur	202	109	62.4%

Keyword Signal In Candidate Rows

LATEST SOURCE KEYWORD	CANDIDATE ROWS	SHARE
Mahadev Mandir	21,582	29.1%
Shankar Mandir	10,189	13.7%
Shiva temple	10,141	13.7%
Mahadev temple	9,954	13.4%
Shiv Mandir	7,936	10.7%
Rudreshwar temple	6,699	9.0%
Lingeshwar temple	2,933	4.0%
Someshwar temple	2,348	3.2%
Vishwanath temple	2,335	3.2%

Keyword signal is based on each candidate row's latest stored source query. It is useful for triage, but not a full attribution history across every rediscovery.

7. Challenges And Research Controls

Challenges Encountered

- Some downloaded files were not what their filenames suggested. One "towns" file was actually Uttar Pradesh sub-district data.
- LGD local body data was valuable, but mixing it into Phase 1.1 would have blurred the district baseline.
- Administrative names vary across sources: spelling variants, renames, parenthetical labels, and legacy district names needed careful alignment.
- Two Assam districts were present in LGD but missing in the database, while one Arunachal pilot row sat outside the LGD district baseline.
- Google Places returns discovery evidence, not a canonical temple registry, so counts must remain uncertainty-aware.
- Repeated discoveries across keywords created heavy duplication, making stable place-ID dedupe essential.

Controls Applied

- Phase 1.1 was explicitly restricted to active district rows.
- LGD district codes were preserved as administrative identifiers.
- Alignment output was written to review CSVs before applying changes.
- Manual exceptions were documented and retained in the audit trail.
- Google Places calls used bounded limits and a narrow field mask.
- Reports state discovery counts rather than final real-world temple counts.

Known Limitations

- Candidate names are Google Places display names, not verified canonical temple names.
- Some real temples may not be in Google Places, may be typed differently, or may not match the selected keywords.
- Some candidates may be non-Shiva temples whose names contain overlapping terms, or Shiva temples whose names do not contain configured terms.
- The report filters candidates by current `source_location_id`. A future candidate-discovery history table is needed for complete multi-query attribution.

8. Path Forward

Immediate Next Research Steps

- Rerun or inspect the one failed district task and document the outcome.
- Sample high, medium, and low confidence candidates across top and low-volume districts.
- Add a candidate discovery history table to retain every query-to-place observation.
- Create a manual review protocol with fields for canonical name, deity verification, source URL, verifier, and confidence notes.
- Keep Phase 1.2 town and ULB expansion as a separate scoped report, not mixed into Phase 1.1.

Suggested External Positioning

Use: An uncertainty-aware district-level discovery baseline of likely Shiva temple candidates in India, generated from official Indian district data and Google Places discovery results.

Avoid: exact number of Shiva temples, official census of temples, complete temple database.

References

Data.gov.in LGD catalog: <https://www.data.gov.in/catalog/local-government-directory-lgd>

Ministry of Panchayati Raj LGD overview: <https://panchayat.gov.in/en/lgd/>

Google Places Text Search documentation: <https://developers.google.com/maps/documentation/places/web-service/text-search>

Local project evidence: `data/prepared_locations_lgd_districts.csv`, `reports/location_verification/*.csv`, `reports/phase_1_1_district_baseline/*.csv`.